

HOPPER INFORMATION SERIES

07 / CUSTOM IMAGE CLASSIFICATION

Build a Teachable Machine model

Train custom image classes in the browser, export TensorFlow.js files and load them into Hopper Studio.

COLLECT

TRAIN

EXPORT

LOAD





Choose classes that answer one mission question

GOOD CLASS SET

RED PAD

BLUE PAD

CLEAR FLOOR

Each frame must belong to one class.

WRITE THE QUESTION FIRST

- Which landing pad fills the camera view?
- Is the floor clear or blocked?
- Which classroom zone is below the drone?

ALWAYS INCLUDE A NEGATIVE

- none, background or clear floor.
- Without it, the model must choose an object class.
- Avoid labels that look nearly identical.
- Use whole-frame categories, not tiny-object location.

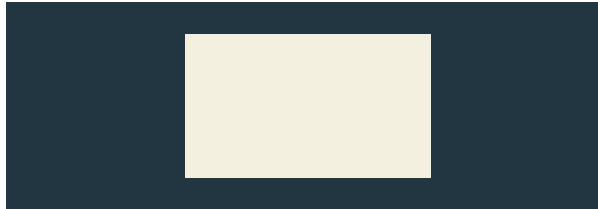


Collect a balanced, realistic dataset

ANGLE



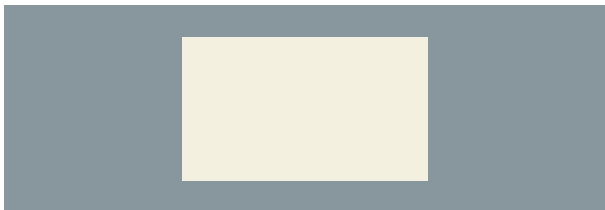
DISTANCE



ROTATION



LIGHT



SHADOW



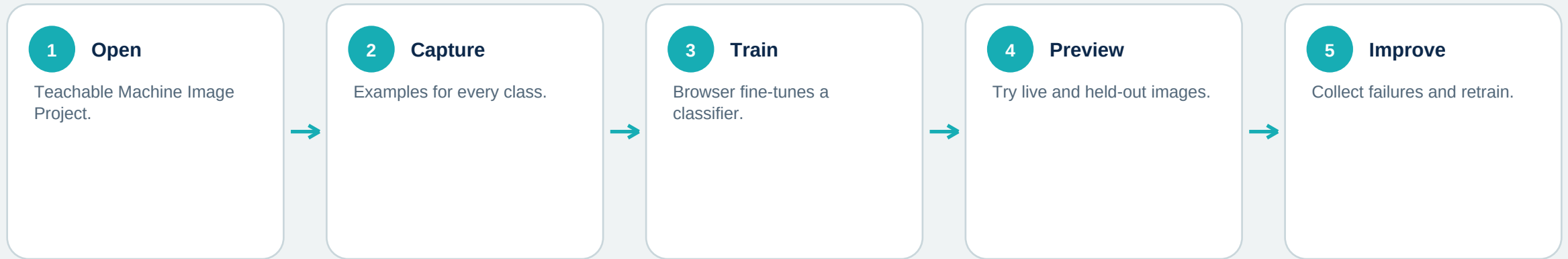
BACKGROUND



Use similar sample counts per class and hold out examples you did not train on.



Train, then validate on unseen views



TRAINING PREVIEW IS NOT ENOUGH

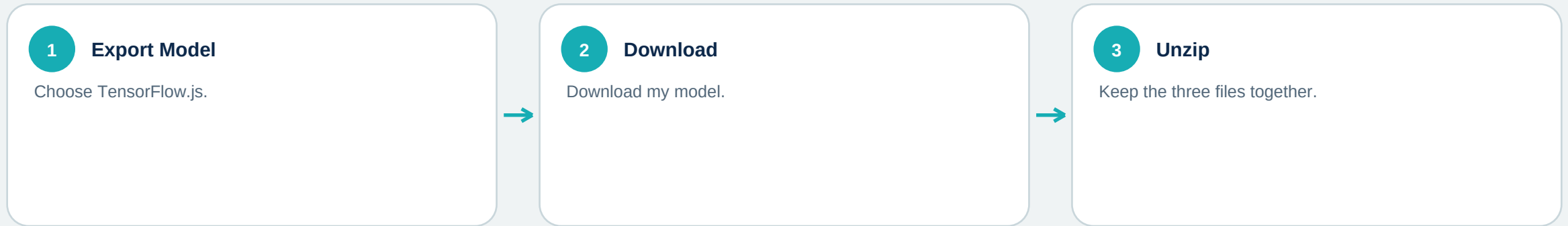
- Test a new room area and a new lighting condition.
- Move the target through the full camera field.
- Test the negative class more often than feels necessary.

TRANSFER LEARNING

A pretrained feature extractor already recognizes useful image patterns. Training adjusts a smaller final classifier for your labels.



Export the standard TensorFlow.js files



hopper-model/

model.json

network structure

weights.bin

learned parameters

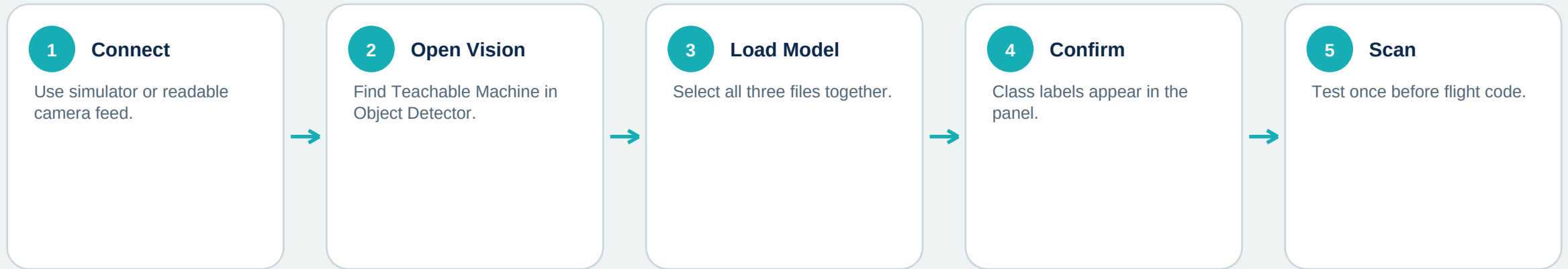
metadata.json

labels + project metadata

Not supported: .tflite, embedded export or only one of the three files.



Load the model into Hopper Studio



IMPORTANT: THE MODEL IS NOT FLASHED INTO THE AIRCRAFT

Hopper Studio loads the files into browser/app memory and runs inference on this computer.

Reload after refresh. The model files are not persisted or uploaded to a cloud service.

MODEL.JSON

WEIGHTS.BIN

METADATA.JSON



Use the custom class in blocks or JavaScript

custom model sees red pad at 75% confidence

CUSTOM CLASS CHECK

```
1  const redPad = await
2    vision.seesCustomLabel("red pad", 0.75);
3  if (redPad) {
4    await drone.hover();
5    await drone.land();
6  }
```

WHAT YOU GET

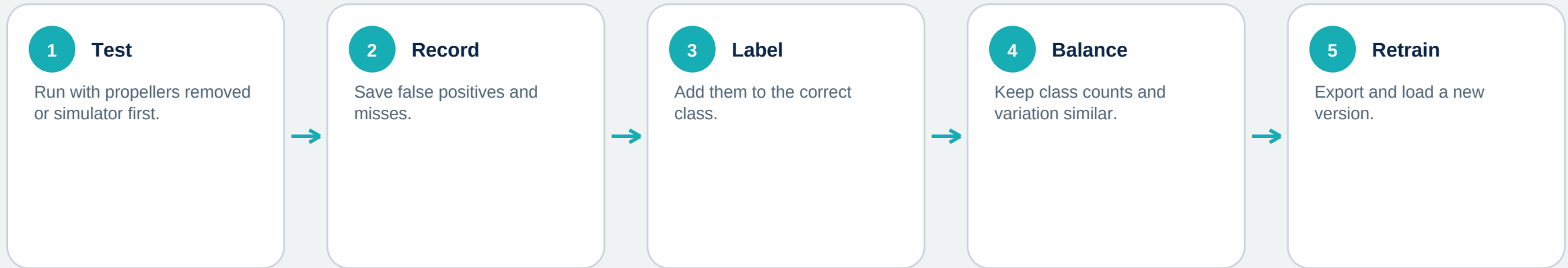
- className + probability for every class.
- Exact class-name match, case-insensitive.
- No bounding box, x/y or multiple-object location.

FRESH SCAN

- seesCustomLabel captures a new frame.
- Confidence in JavaScript is 0..1.
- Scanning includes animation + inference time.
- Do not use it as a high-rate control sensor.



Treat every failure as new training data



FLIGHT ACCEPTANCE TEST

- Target class succeeds across distance/rotation.
- Negative class rejects empty floor.
- No single lighting condition dominates.
- Threshold chosen from measured errors.

SOURCES AND FURTHER READING

Google Teachable Machine

<https://teachablemachine.withgoogle.com/train>

TensorFlow.js transfer learning overview

<https://codelabs.developers.google.com/tensorflowjs-transfer-learning-teachable-machine>

Hopper Studio custom-model implementation

[lib/vision.ts](#) and [README.md](#)